tion from statistical metal weight prediction methods. Thus, the weights for these components were developed using available statistical weight equations and preliminary design methods with corrections for advanced composite material technical applications as indicated earlier. The resulting aircraft weight breakdown is presented in Table 3.

TABLE 3. TOTAL AIRCRAFT WEIGHT BREAKDOWN

	WT-LB	PERCENT
PROPULSION	25,740	25.6
STRUCTURE	24,715	24.5
EQUIPMENT	13,260	13.2
USEFUL LOAD	22,230	22.1
FUEL (VTOL)	14,735	14.6
TOTAL	100,680	100.0

Review of the weight breakdown of Table 3 indicates that the propulsion and structure represent approximately 40 and 39 percent of the empty weight of the vehicle, respectively. The fuel for the VTOL mission represents only 14.6 percent of the total VTOL takeoff weight, this is due to the short trip distance requirements of the aircraft for operations in the VTOL mode.

The center of gravity and moment of inertia characteristics of the vehicle are illustrated in figure 18. The data show that the aircraft has a reasonably small CG shift with fuel load. The maximum forward and aft CG travel due to a ± 5% cabin length shift of the payload are indicated. The yaw axis has the highest moment of inertia, followed by pitch. The roll axis has the lowest moment of inertia -- primarily because the lift/cruise propulsion system does not contribute significantly to the inertia of this axis.

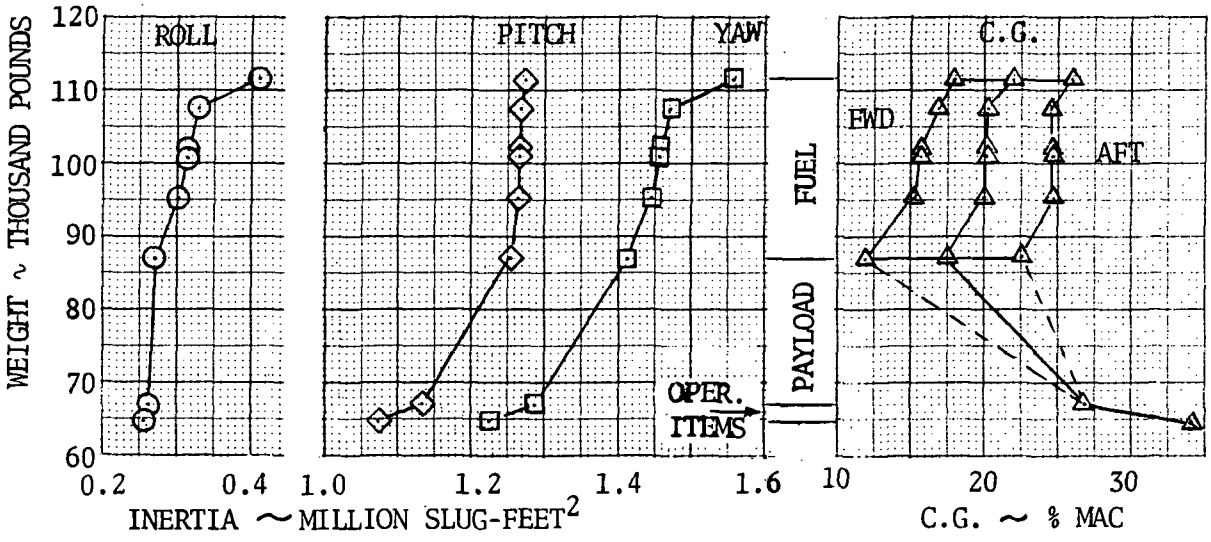


Figure 18. Inertia and Center of Gravity Characteristics